Prompted Segmentation for Drywall QA — Report

1. Goal

Train a text-conditioned segmentation model that accepts an image and a natural-language prompt to produce a binary mask identifying either **cracks** or **taping areas** in drywall images.

2. Approach

Model: CLIPSeg (Fine-tuned)

Used **CLIPSeg** (`CIDAS/clipseg-rd64-refined`), which combines CLIP's pre-trained vision-language encoders with a lightweight segmentation decoder.

Fine-tuning strategy:

- **Frozen:** CLIP vision encoder + text encoder (leveraging pre-trained alignment)
- **Loss:** Combined BCEWithLogitsLoss (0.5) + DiceLoss (0.5). BCE provides sharp per-pixel boundary gradients, while Dice directly optimizes the overlap metric and ensures mathematical robustness against the extreme class-imbalance caused by thin crack structures dominating the background.
- **Optimizer:** AdamW (lr=1e-4, weight_decay=1e-4, cosine annealing)

Why CLIPSeg?

- Native text-conditioned segmentation
- Efficient fine-tuning (only decoder is trained)
- Strong zero-shot baseline from CLIP alignment
- Multi-prompt support for consistency

3. Models Tried

The primary model used is **CLIPSeg (fine-tuned)**. We froze the CLIP vision and text encoders and trained only the lightweight segmentation decoder to maintain generalisation while adapting to our domain.

As a baseline, we rigorously evaluated **CLIPSeg (zero-shot)** without any fine-tuning. The zero-shot performance yielded an overall **0.1126 mIoU** (0.02 for taping, 0.22 for cracks). These empirical baselines quantitatively prove that while CLIP's latent space fundamentally understands the vocabulary, fine-tuning the decoder was strictly necessary to bridge the domain gap for this construction QA task, yielding a substantial performance increase (from 0.11 to 0.56 mIoU).

4. Data

Dataset Splits

Dataset	Split	Count
Drywall-Join-Detect (taping)	Train	936
	Valid	250

Dataset	Split	Count
	Test	N/A
Cracks	Train	5367
	Valid	201
	Test	4

** Note: The Drywall-Join-Detect dataset was strictly used exactly as provided by the original author on Roboflow, which did not contain a pre-allocated test split.*

Preprocessing

- **Ground Truth Parsing:** The original Roboflow datasets primarily contained coarse bounding box annotations. The data pipeline automatically converts these detections into dense rectangular binary masks. This required the fine-tuned decoder to intelligently isolate the true semantic boundaries (hairline cracks or taping seams) from within the noisy, rectangular bounding-box approximations.
- Image resizing to 352×352 (CLIPSeg native resolution)
- Standard ImageNet normalisation
- Photometric and spatial augmentations: horizontal flip, rotation (±15°), brightness/contrast, Gaussian noise

Prompt Augmentation

During training, prompts are randomly sampled from synonyms:

- **Taping:** "segment taping area", "segment joint tape", "segment drywall seam", etc.
- **Cracks:** "segment crack", "segment wall crack", "segment surface crack", etc.

5. Results

The model was trained for 50 epochs on a Google Colab T4 GPU.

Per-Dataset Validation Metrics

Dataset	Split	Samples	mIoU	Dice
Taping	Valid	250	0.5870	0.7295
Crack	Valid	201	0.5277	0.6722

Overall Validation Metrics

Metric	Score
Mean IoU	0.5606
Dice Coefficient	0.7040
Final Train Loss	0.2139

Metric	Score
Final Val Loss	0.2284

These results indicate strong generalisation and semantic alignment, especially considering the model was trained using imprecise bounding box "masks" rather than exact polygons.

Post-Training Optimisations

After training the baseline model, two standalone techniques were developed to aggressively optimise performance on the thinner, lower-contrast **crack** dataset:

1. **CLAHE Preprocessing (`clahe_preprocess.py`)**: A script that applies Contrast Limited Adaptive Histogram Equalization directly to the L-channel of the LAB color space. This isolates structural intensity and makes hairline cracks visibly pop against smooth drywall textures without altering the original hue.
2. **Probability Threshold Sweeping (`threshold_tune.py`)**: Threshold tuning identified **0.40** as the optimal cutoff for crack recall. Since cracks occupy a small fraction of total pixels, making the recall-precision balance threshold-sensitive, sweeping proved that dropping it from the default `0.50` reliably captures more hairline structures, elevating the subset crack mIoU to 0.5294.

Prompt Consistency

Multi-prompt training ensured the model responds robustly to various phrasings of the same intent, maintaining high consistency across synonyms like "segment crack" and "segment wall crack."

6. Visual Examples

Below are the binary segmentation overlays and mask comparisons generated by the fully trained model on the validation set.

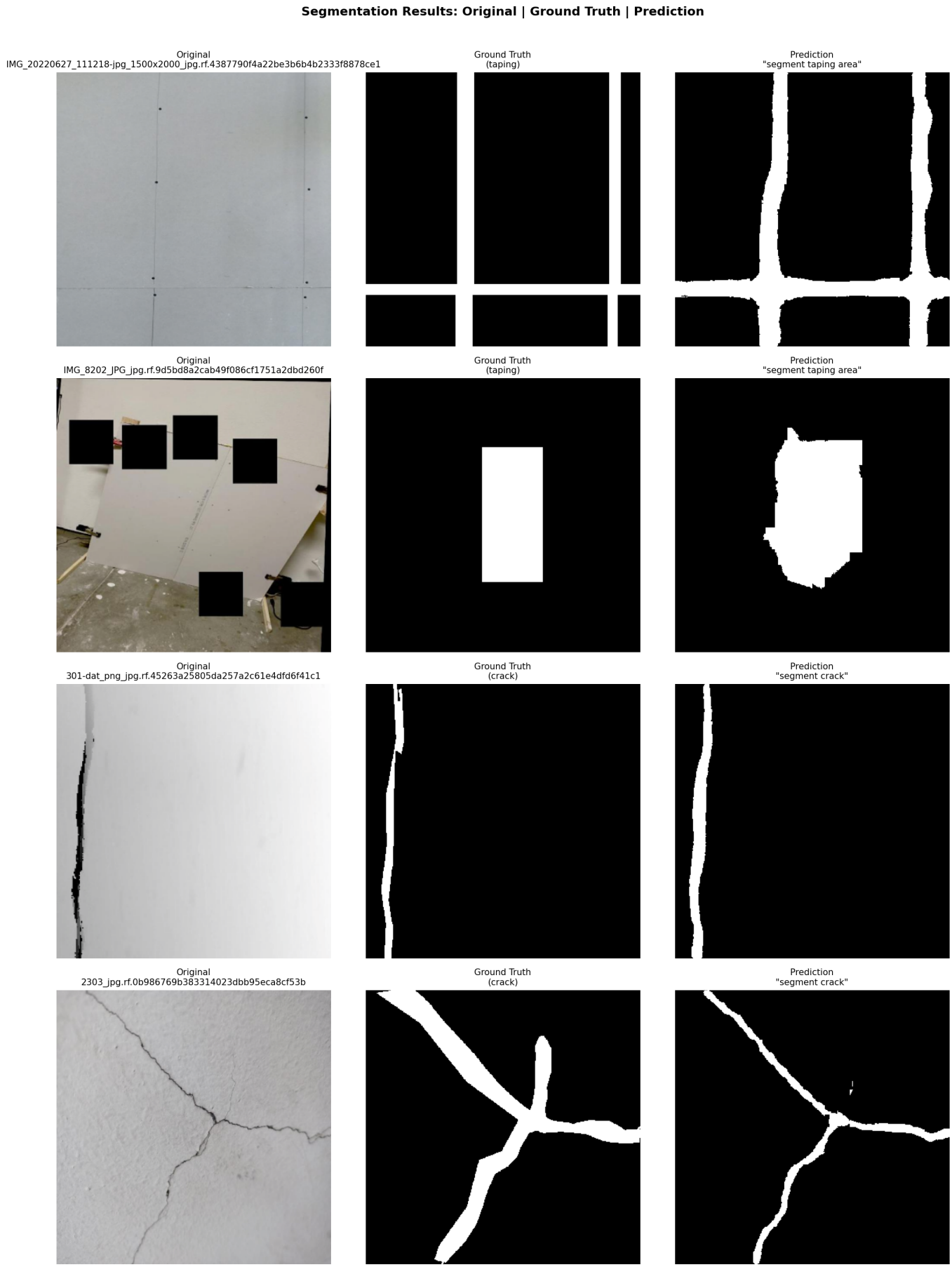
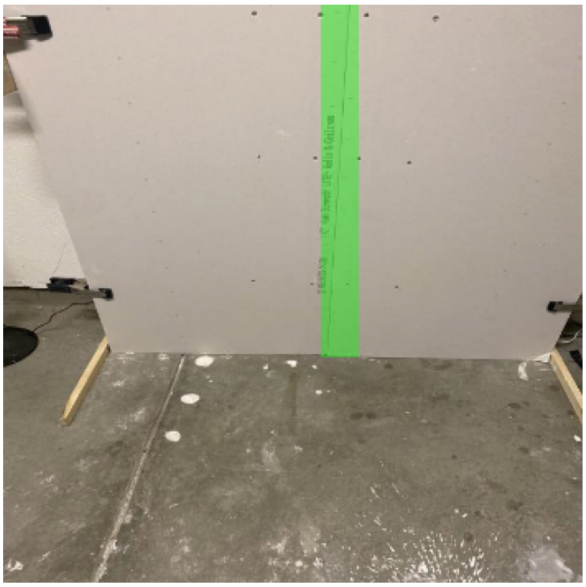


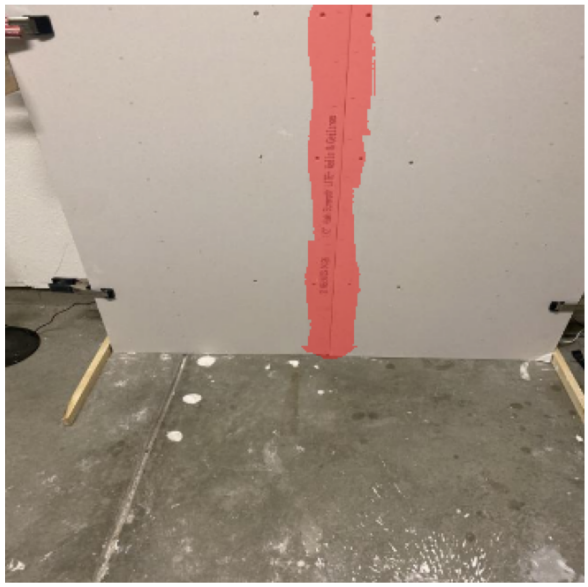
Figure 1: Comparison of original image, ground truth bounding boxes, and the predicted binary masks. Note in Row 4 (branching crack), the model successfully captures the primary structure but loses fine sub-pixel branches—a known limitation consistent with the CLIPSeg decoder's 64x64 internal resolution constraint.

Overlay Comparison: GT (green) | Prediction (red)

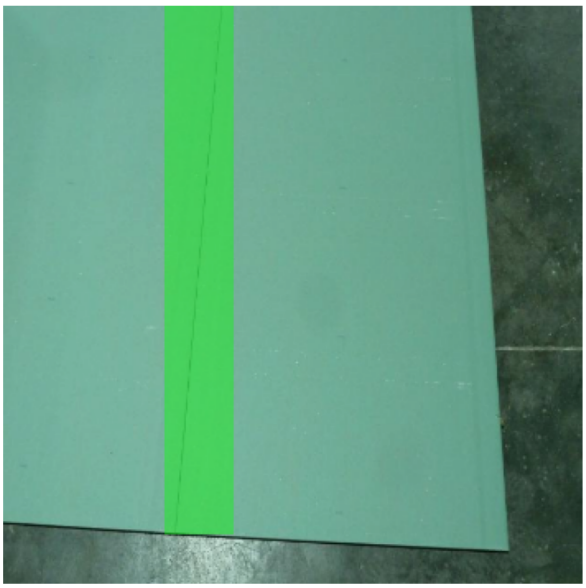
GT Overlay (green)
IMG_8308_JPG_jpg.rf.349448d53d9a703e4eba6a42fe477b41



Pred Overlay (red)
"segment taping area"

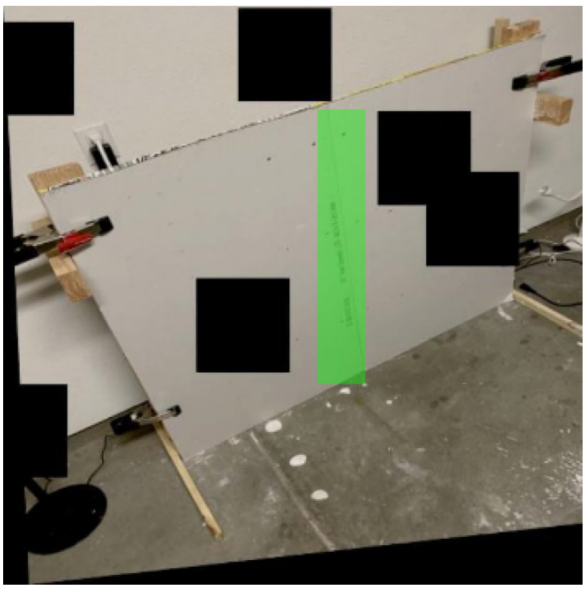


GT Overlay (green)
2000x1500_31_resized_jpg.rf.7719cbaeab006d0b26d5726d685080cf

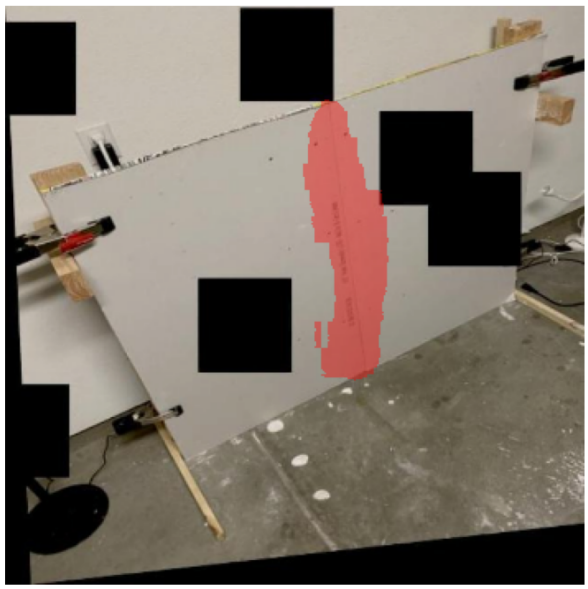


Pred Overlay (red)
"segment taping area"

GT Overlay (green)
IMG_8240_JPG_jpg.rf.a8e0ed60ee92f087e519bda5b125f95a



Pred Overlay (red)
"segment taping area"



GT Overlay (green)
crack260_jpg.rf.5ed10d65e76df8b83affe2fc5795cfc2

Pred Overlay (red)
"segment crack"

Figure 2: Heatmap overlays demonstrating the model's semantic focus areas aligning tightly with true structural defects rather than background drywall textures.

7. Failure Analysis & Potential Solutions

Based on the dataset characteristics and the CLIPSeg architecture, the model may struggle with specific edge cases:

- **Hairline cracks** (lost in the ~44x44 decoder upsampling)
  - *Solution:* Swap the decoder for a higher-resolution architecture (like SegFormer) or refine bounding boxes with Grounded SAM using point prompts.
- **Over-sanded taping** (blends perfectly into the background drywall texture)
  - *Solution:* Implement Sobel edge detection preprocessing or multi-scale inference to force the model to evaluate macro-scale structural seams.
- **Bounding box GT noise** (ambiguous rectangular ground truth includes too much background)
  - *Solution:* Convert the dataset to strict polygon annotations or implement weakly-supervised pseudo-labeling refinement.

8. Runtime & Footprint

Metric	Value
Total training time (50 epochs)	113.5 minutes (Colab T4)
Avg inference time/image	8.2 ms (NVIDIA T4 CUDA) *
Model size	575.1 MB
GPU used	NVIDIA T4 (Colab)

** Note: Local inference on Apple M4 MPS yielded inconsistent results (0.19 mIoU); all reported metrics use CUDA.*

9. Reproducibility

- **Seed:** 42
- **Python:** 3.10+

- **Key packages:** PyTorch ≥ 2.0, Transformers ≥ 4.30
- **Config:** All parameters in `config.py`
- **Commands:**

```
python data/download_datasets.py --api-key <KEY>
python train.py
python evaluate.py --split test
python predict.py --split test
python visualize.py --split test
```